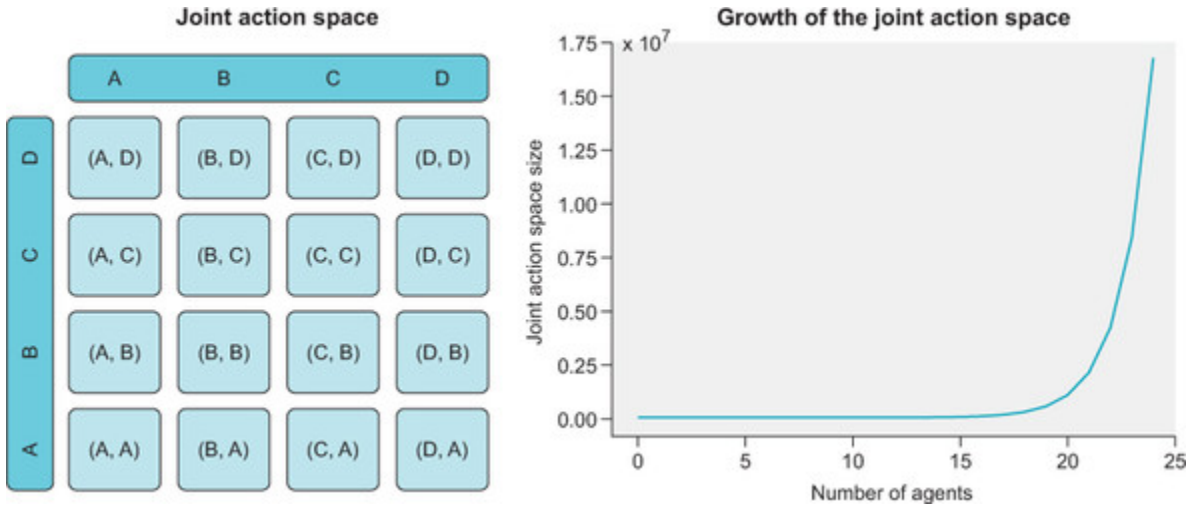
Gridworld—then we have to use a $4^2 = 16$ length one-hot vector instead of just a 4 length vector. This is because there are 16 different possible combinations of actions between two agents with 4 actions each: [Agent 1: Action 1, Agent 2: Action 4], [Agent 1: Action 3, Agent 2: Action 3] and so on (see figure 9.5).

Figure 9.5. If each agent has an action space of size 4 (i.e., it is represented by a 4 element one-hot vector), the joint action space of two agents is 42 = 16, or 4N where N is the number of agents. This means the growth of the joint action space is exponential in the number of agents. The figure on the right shows the joint action space size for agents with individual action spaces of size 2. Even with just 25 agents, the joint action space becomes a 33,554,432 element one-hot vector, which is computationally impractical to work with.



If we want to model the joint action of 3 agents, we have to use a $4^3 = 64$ length vector. So, in general for Gridworld, we have to use a 4^N length vector, where N is the number of agents. For any environment, the size of the joint action vector will be $|A|^N$ where $|A|$ refers to the size of the action space (i.e., the number of discrete actions). That is an exponentially growing vector in the number of agents, and this is impractical and intractable for any significant number of agents. Exponential growth is always a bad thing, since it means your algorithm can't scale. This exponentially large joint action space is the main new complication that *multi-agent reinforcement learning* (MARL) brings, and it is the problem we'll spend this chapter solving.